

The Relational Spin-Closure Hypothesis

From Euclidean Three-Space to Conditional Lorentzian Spin Geometry

Edited by Jan Seeck

Abstract

We study a reconstruction problem whose primitive local input is only the standard positive-definite Euclidean space $V = \mathbb{R}^3$. From this datum one obtains the real spin factor $\mathcal{S} = \mathbb{R} \oplus V$, its quadratic form $N(a, v) = a^2 - \langle v, v \rangle$ of signature $(1, 3)$, the real Clifford algebra $\text{Cl}(V, q)$, an intrinsic Spin group G_{Spin} , and a surjective central double cover

$$1 \longrightarrow \{\pm 1\} \longrightarrow G_{\text{Spin}} \xrightarrow{\rho} G_{\text{Lor}} \longrightarrow 1$$

of an intrinsically defined proper cone-preserving Lorentz group. In parallel, the Lorentz-skew endomorphisms of $\mathcal{S}$ form a six-dimensional Lie algebra $\mathfrak{g}_B$ linearly equivalent to $\Lambda^2 \mathcal{S}$.

The passage from this local algebra to a global base is not automatic. We exhibit a theorem-level non-derivability result: identical local pieces can be assembled by different incidence data into non-homeomorphic quotient spaces. We therefore isolate explicit base-gluing data and prove that, under a closed-gluing-graph condition, countability, and a sufficient smoothness condition, the quotient is a Hausdorff second-countable smooth four-manifold M .

A second non-derivability result separates the internal Lorentz transition system from the genuine tangent transition system $D\phi_{ij}$. Their identification requires an additional regular solder datum. Conditional on such a solder, the intrinsic form on $\mathcal{S}$ descends to a smooth, chart-independent, nondegenerate Lorentzian $(0, 2)$ -tensor of signature $(1, 3)$ on TM ; a project-native orientation reduction and a coherent future-cone reduction follow. The same solder identifies the tangent Lorentz cocycle with the projection of a native Spin cocycle, forcing the project-native orientation and Spin-lifting obstructions to vanish. No identification of these project-native obstructions with $w_1(TM)$ or $w_2(TM)$ is made.

Finally, on one fixed periodic control, we analyse a selected one-parameter Spin transition family generated by a chosen Clifford direction. Every finite parameter value is regularly admissible; all parameters lie in one full Spin Čech gauge orbit; and the complete regular-solder solution spaces at two parameter values are explicitly bijective with identical induced tangent metrics under the correspondence. This is a scoped negative control on a transition-cocycle deformation, not a general deformation classification and not a construction of connection, holonomy, curvature, or dynamics.

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1 Introduction

The construction of a Lorentzian spin geometry is usually formulated after a smooth manifold, a Lorentz metric, and the associated frame bundle have already been supplied; the Spin structure then appears as a lift of an orthonormal frame bundle subject to a topological obstruction [6, 8, 11]. Here the order of dependence is deliberately reversed. The starting object is only

$$V = \mathbb{R}^3, \quad \langle u, v \rangle = u_1 v_1 + u_2 v_2 + u_3 v_3,$$

with no four-dimensional base space, tangent bundle, Lorentz form, orientation, Spin bundle, connection, or curvature in the input.

There are then three logically separate reconstruction questions.

First, what local Lorentzian and intrinsic Spin-algebraic structures can be constructed from the local Euclidean algebra? Second, what information must be added before local copies of the resulting model can be assembled into a global smooth manifold? Third, after such a manifold has been obtained, what additional datum is required to identify the internal Lorentz system with its genuine tangent geometry?

The separation of these questions is essential. A local model is not a manifold; fibre transition data do not determine which local base points are to be identified; and an internal Lorentz cocycle is not, without further structure, the derivative cocycle of a manifold atlas. We make each of these boundaries explicit and place the positive reconstruction theorems immediately after the relevant missing datum.

The local algebra belongs to the standard circle of ideas around real Clifford algebras, Spin groups, spin factors and Lorentz cones [1, 6, 4, 2]. The global steps use standard quotient, manifold, bundle and Čech ideas [7, 3, 11, 5], but the point of the present organization is not to assume the usual global Lorentzian package in advance. Instead, the precise dependency chain is

$$\begin{array}{c}
 (\mathbb{R}^3, \langle \cdot, \cdot \rangle) \\
 \Downarrow \\
 (\mathcal{S}, N, B, \text{Cl}_3, G_{\text{Spin}}, \rho, \mathfrak{g}_B) \\
 + \text{base incidence and gluing} \\
 \Downarrow \\
 \text{topological four-manifold candidate} \\
 + \text{closed gluing graph} + \text{countable index} + \text{smooth gluing} \\
 \Downarrow \\
 M^4, \quad TM, \quad D\phi_{ij} \\
 + \text{regular solder} \\
 \Downarrow \\
 (TM, g, \text{orientation reduction, future-cone reduction, native Spin reconvergence}).
 \end{array} \tag{1.1}$$

The boxes marked by plus signs are genuine gates: their inhabitation is not derived from the preceding local algebra.

The following six statements organize the paper.

Theorem 1.1 (Local reconstruction). *From the specified Euclidean datum $V = \mathbb{R}^3$, the construction produces:*

1. *a four-dimensional real spin factor $\mathcal{S} = \mathbb{R} \oplus V$ with Lorentz form of signature (1, 3);*
2. *the real Clifford algebra $\text{Cl}_3 = \text{Cl}(V, q)$, with the paravector embedding of $\mathcal{S}$;*
3. *an intrinsic Spin group G_{Spin} , an intrinsic proper cone-preserving Lorentz group G_{Lor} , and a continuous surjective homomorphism $\rho : G_{\text{Spin}} \rightarrow G_{\text{Lor}}$ whose central kernel is exactly $\{\pm 1\}$, together with continuous local sections;*

4. a six-dimensional Lorentz-skew Lie algebra $\mathfrak{g}_B$ and a linear equivalence $\Lambda^2 \mathcal{S} \simeq \mathfrak{g}_B$.

Theorem 1.2 (Conditional smooth globalization). *The local model does not determine a global base. Given explicit compatible base-incidence/gluing data B , a closed gluing graph, a countable index set, and the sufficient condition that the base transition maps are C^∞ , the quotient M_B is a Hausdorff second-countable smooth four-manifold modelled on $\mathcal{S}$.*

Theorem 1.3 (Tangent–internal independence). *For the reconstructed smooth manifold, the tangent transition system is the derivative system $D\phi_{ij}$. A coherent native Spin cocycle gives, independently, an internal Lorentz transition system $\rho(\tilde{g}_{ij})$. Smooth base gluing together with coherent native Spin data does not force these two systems to agree.*

Theorem 1.4 (Regular solder closure). *A regular solder is an additional smooth local family of fibrewise linear equivalences satisfying the exact overlap intertwining law between $D\phi_{ij}$ and $\rho(\tilde{g}_{ij})$. Conditional on its existence, it yields a bundle-level equivalence certificate between the internal Lorentz bundle and TM , with continuous total-space maps in both directions and smooth local representatives. It also induces a smooth Lorentzian tangent metric of signature $(1, 3)$.*

Theorem 1.5 (Global reconvergence). *For every regular solder, the tangent Lorentz transition system in solder frames equals the projected native Spin cocycle. Consequently the project-native determinant obstruction is a coboundary and the project-native fixed-cover Spin-lifting obstruction is trivial. A regular solder also yields a coherent future-cone reduction.*

Theorem 1.6 (Selected-control gauge absorption). *On the fixed periodic control and for the selected one-parameter native Spin transition family considered in section 11, every finite parameter value admits a regular solder. Any two parameter values define full-Spin-Čech-gauge-equivalent cocycles. Their complete regular-solder solution spaces are explicitly bijective, and corresponding solders induce the same tangent Lorentz metric.*

The hypotheses in theorems 1.2 and 1.4 are not cosmetic. They are exactly where the construction ceases to be intrinsic to the initial Euclidean local data. In particular, this paper does *not* prove an implication

$$\mathbb{R}^3 \implies \text{a canonical spacetime.}$$

It proves the local implication in theorem 1.1, identifies the additional global data required to continue, and then proves what follows from those data.

2 The Euclidean primitive and the Lorentzian spin factor

2.1 Primitive input

Throughout the local part of the paper,

$$V := \mathbb{R}^3$$

carries only its standard positive-definite bilinear form

$$\langle u, v \rangle := u_1 v_1 + u_2 v_2 + u_3 v_3.$$

Set

$$q_3(v) := \langle v, v \rangle.$$

No Lorentzian structure has yet been introduced.

Definition 2.1 (Real spin factor). Define

$$\mathcal{S} := \mathbb{R} \oplus V$$

and, for $x = (a, u)$, $y = (b, v)$, define

$$x \circ y := (ab + \langle u, v \rangle, av + bu).$$

The distinguished unit is

$$\mathbf{1}_{\mathcal{S}} := (1, 0).$$

The product is bilinear and commutative. As an auxiliary local identity for this chosen spin-factor construction one also has the Jordan identity

$$(x \circ y) \circ (x \circ x) = x \circ (y \circ (x \circ x)).$$

Thus $(\mathcal{S}, \circ, \mathbf{1}_{\mathcal{S}})$ is the rank-two real spin factor; compare the standard Jordan-algebra treatment in [4, 2]. This local identity is supporting algebra, not an additional global reconstruction claim.

Definition 2.2 (Intrinsic quadratic and bilinear forms). For $x = (a, u) \in \mathcal{S}$, set

$$N(x) := a^2 - \langle u, u \rangle.$$

Its polarization is

$$B(x, y) := \frac{N(x+y) - N(x) - N(y)}{2} = ab - \langle u, v \rangle$$

for $y = (b, v)$.

Proposition 2.3 (Lorentz signature). *The form B is symmetric and nondegenerate, and has signature $(1, 3)$. In the standard basis*

$$\mathbf{1}_{\mathcal{S}}, \quad (0, \varepsilon_0), \quad (0, \varepsilon_1), \quad (0, \varepsilon_2),$$

its Gram matrix is

$$\text{diag}(1, -1, -1, -1).$$

Proof. The displayed formula $B((a, u), (b, v)) = ab - \langle u, v \rangle$ gives the Gram matrix directly. Since the Euclidean form on V is positive definite, the first basis direction is positive and the remaining three are negative. The determinant is nonzero, hence B is nondegenerate. $\square$

A second auxiliary local identity of the spin-factor construction is the Cayley–Hamilton identity

$$x \circ x - \operatorname{tr}_{\mathcal{S}}(x)x + N(x)\mathbf{1}_{\mathcal{S}} = 0, \quad \operatorname{tr}_{\mathcal{S}}(a, u) = 2a. \quad (2.1)$$

The square cone

$$C_{\mathcal{S}} := \{x \in \mathcal{S} : \exists y \in \mathcal{S}, x = y \circ y\} \quad (2.2)$$

provides an intrinsic time-cone structure at the level of the local model.

Definition 2.4 (Intrinsic Lorentz group). Let G_{Lor} be the group of real-linear self-equivalences $F : \mathcal{S} \rightarrow \mathcal{S}$ satisfying

$$N(Fx) = N(x), \quad x \in C_{\mathcal{S}} \iff Fx \in C_{\mathcal{S}}, \quad \det F = 1. \quad (2.3)$$

The definition is internal to $(\mathcal{S}, N, C_{\mathcal{S}})$. In particular, no matrix realization over $\mathbb{C}$, no pre-existing Lorentz group on a spacetime, and no frame bundle enters the construction.

Boundary statement 2.5 (Local means local). At this point $\mathcal{S}$ is a four-dimensional real vector space with an intrinsic Lorentz form. It is not a tangent space and there is no manifold, chart, atlas, base point, or covering space in the construction.

3 Clifford envelope and the intrinsic Spin double cover

3.1 The Euclidean Clifford algebra

Let

$$\text{Cl}_3 := \text{Cl}(V, q_3)$$

be the real Clifford algebra characterized by the relation

$$\iota(v)^2 = q_3(v)\mathbf{1}. \quad (3.1)$$

For the standard Euclidean basis $\varepsilon_0, \varepsilon_1, \varepsilon_2$ of V , write $\gamma_i := \iota(\varepsilon_i)$. Then

$$\gamma_i^2 = 1, \quad \gamma_i \gamma_j = -\gamma_j \gamma_i \quad (i \neq j). \quad (3.2)$$

The signs are those of the *positive* Euclidean quadratic form on V ; the Lorentzian signature has already appeared in the distinct four-dimensional spin factor $\mathcal{S}$.

Definition 3.1 (Paravector embedding). Define

$$A : \mathcal{S} \longrightarrow \text{Cl}_3, \quad A(a, v) := a\mathbf{1} + \iota(v). \quad (3.3)$$

Let c denote the Clifford conjugation used in the construction. Its restriction to paravectors changes the sign of the vector part. The key identity is

$$A(x)c(A(x)) = N(x)\mathbf{1}. \quad (3.4)$$

Proposition 3.2 (Jordan–Clifford compatibility). *For all $x, y \in \mathcal{S}$,*

$$A(x \circ y) = \frac{1}{2}(A(x)A(y) + A(y)A(x)). \quad (3.5)$$

Moreover, the image $A(\mathcal{S})$ generates Cl_3 as a real algebra.

Proof. Expand the product using

$$\iota(u)\iota(v) + \iota(v)\iota(u) = 2\langle u, v \rangle \mathbf{1}.$$

The scalar part becomes $2(ab + \langle u, v \rangle)$, while the vector part becomes $2(av + bu)$, proving (3.5). Since $A(0, v) = \iota(v)$, the image contains every Clifford generator; the universal generation property of the Clifford algebra then gives the second assertion. $\square$

As standard background for the Matlab `CliffordAlgebra` construction used here, Cl_3 carries the usual universal property: any real algebra receiving a linear map $f : V \rightarrow \mathcal{A}$ with $f(v)^2 = q_3(v)\mathbf{1}$ receives a unique algebra homomorphism from Cl_3 . This is a property of the standard Clifford-algebra construction rather than an additional registered global claim of the present reconstruction; see [1, 6].

3.2 Intrinsic Spin elements

Definition 3.3 (Spin element and Spin group). An element $g \in \text{Cl}_3$ is a Spin element when

$$g c(g) = c(g) g = 1. \quad (3.6)$$

The corresponding units form a subgroup

$$G_{\text{Spin}} \subset \text{Cl}_3^\times. \quad (3.7)$$

The Clifford action used in the construction preserves the paravector subspace and defines a real-linear action

$$\text{spinLor}(g) : \mathcal{S} \longrightarrow \mathcal{S}.$$

For every $g \in G_{\text{Spin}}$, this action belongs to G_{Lor} . Hence

$$\rho : G_{\text{Spin}} \longrightarrow G_{\text{Lor}}, \quad \rho(g) = \text{spinLor}(g), \quad (3.8)$$

is a group homomorphism.

Theorem 3.4 (Intrinsic double cover). *The homomorphism ρ is surjective and*

$$\ker \rho = \{1, -1\}. \quad (3.9)$$

The kernel is central and has two elements. With the topologies inherited from the finite-dimensional Clifford model and the induced topology on G_{Lor} , both groups are Hausdorff topological groups, ρ is continuous, the kernel is discrete, and every $k \in G_{\text{Lor}}$ has an open neighbourhood U admitting a continuous local section

$$s : U \longrightarrow G_{\text{Spin}}, \quad \rho \circ s = \text{id}_U. \quad (3.10)$$

Proof. We separate surjectivity, the kernel calculation, and the local-section statement.

Surjectivity. Let $F \in G_{\text{Lor}}$ and put $p = F(1_{\mathcal{S}})$. Because F preserves both N and the intrinsic square cone,

$$N(p) = 1, \quad p \in C_{\mathcal{S}}, \quad p_0 \geq 0.$$

Write $p = (a, u)$. From $a^2 - \langle u, u \rangle = 1$ and $a \geq 0$ one has $a \geq 1$. Hence the forward spin-factor square root

$$r = \sqrt[4]{p} := \left(\sqrt{\frac{a+1}{2}}, \frac{u}{2\sqrt{(a+1)/2}} \right)$$

is well defined and satisfies $r \circ r = p$ and $N(r) = 1$. Set $\beta = A(r) \in \text{Cl}_3$. The Clifford norm identity makes β a Spin element, and the Jordan–Clifford compatibility gives

$$\rho(\beta)(1_{\mathcal{S}}) = p.$$

Consequently

$$G := \rho(\beta)^{-1}F$$

fixes $1_{\mathcal{S}}$. The stabilizer of $1_{\mathcal{S}}$ acts on the spatial summand as an element $R \in SO(3)$. By Cartan–Dieudonné, R is a product of hyperplane reflections in unit spatial vectors. For a unit vector u , the Clifford generator $\iota(u)$ acts on the spatial summand as minus the corresponding hyperplane reflection. Since $\det R = 1$, an even number of reflections occurs; the accumulated minus sign therefore disappears, and the product of the corresponding Clifford generators is a Spin element γ with $\rho(\gamma) = G$. Thus $\rho(\beta\gamma) = F$.

Kernel. Suppose $g \in G_{\text{Spin}}$ acts trivially on every paravector. Evaluating at $1_{\mathcal{S}}$ first shows that Clifford reversion and Clifford conjugation agree on g . The trivial-action equation then implies

$$gA(x) = A(x)g \quad \text{for every } x \in \mathcal{S}.$$

Because the paravector image generates Cl_3 as a real algebra, g commutes with all of Cl_3 . The intrinsically computed centre is $\mathbb{R} \oplus \mathbb{R}\omega$, where $\omega = \gamma_0\gamma_1\gamma_2$ and $\omega^2 = -1$. Hence $g = a + b\omega$. For central elements Clifford conjugation fixes g , so the Spin normalization gives

$$(a + b\omega)^2 = 1 \implies a^2 - b^2 = 1, \quad 2ab = 0.$$

The first equation excludes $a = 0$; therefore $b = 0$ and $a = \pm 1$. Conversely, ± 1 act trivially. Thus $\ker \rho = \{\pm 1\}$; centrality and cardinality two follow immediately.

Continuous local sections. The construction is also intrinsic. Define the Clifford contraction

$$\mathcal{F}(y) := y + \sum_{i=0}^2 \gamma_i y \gamma_i.$$

In the eight-monomial Clifford frame this kills the six noncentral monomials and multiplies the two central ones by 4, so $\mathcal{F}(y) \in \mathbb{R} \oplus \mathbb{R}\omega$. Next define the reconstruction map

$$\mathcal{R}(F) := A(F1_S) + \sum_{i=0}^2 A(F(0, \varepsilon_i)) \gamma_i.$$

For a Spin element g , direct Clifford multiplication gives

$$\mathcal{R}(\rho(g)) = g \mathcal{F}(\text{rev } g),$$

so reconstruction recovers g up to a central factor. Identify the centre $\mathbb{R} \oplus \mathbb{R}\omega$ with $\mathbb{C}$ by the real-algebra embedding $\text{cx} : \mathbb{C} \rightarrow \text{Cl}_3$ determined by $\text{cx}(i) = \omega$, and choose the linear retraction cxRetract used by the reconstruction. Define the continuous discriminant

$$\Delta(F) := \text{cxRetract}(\mathcal{R}(F) c(\mathcal{R}(F))).$$

The key identity is

$$\mathcal{R}(\rho(g)) = \text{cx}(w)g \implies \Delta(\rho(g)) = w^2.$$

Thus the central ambiguity is visible only through a square. At the identity, $\Delta(1) = 16$. On the open neighbourhood where $\Delta(F)$ lies in a slit plane avoiding the branch cut, the principal complex square root is continuous. Dividing the reconstructed Clifford element by this square root leaves exactly the two choices differing by $\{\pm 1\}$, and yields a continuous Spin-valued section s with $\rho(s(F)) = F$ and $s(1) = 1$. Finally, for arbitrary $k \in G_{\text{Lor}}$, choose one lift u_k of k and left-translate the identity neighbourhood and its section. This produces an open neighbourhood of k with a continuous local section. A different choice of u_k changes that section only by the central kernel element -1 . $\square$

Thus one obtains the exact central extension

$$1 \longrightarrow \{\pm 1\} \longrightarrow G_{\text{Spin}} \xrightarrow{\rho} G_{\text{Lor}} \longrightarrow 1. \quad (3.11)$$

A classical complex-matrix description may be used as a comparison endpoint, but it is not an input to theorem 3.4. In particular, the construction does not define G_{Spin} by first declaring it to be $\text{SL}(2, \mathbb{C})$.

4 The intrinsic infinitesimal Lorentz sector

The local group construction has a parallel infinitesimal object that can be defined without first making G_{Spin} into a Lie group.

Definition 4.1 (Lorentz-skew endomorphisms). Let

$$\mathfrak{g}_B := \{T \in \text{End}_{\mathbb{R}}(\mathcal{S}) : B(Tx, y) + B(x, Ty) = 0 \text{ for all } x, y \in \mathcal{S}\}. \quad (4.1)$$

For $u, v \in \mathcal{S}$, define

$$K_{u,v}(x) := B(v, x)u - B(u, x)v. \quad (4.2)$$

Then $K_{u,v} \in \mathfrak{g}_B$, the map is alternating in u, v , and it therefore induces a linear map

$$\Phi : \Lambda^2 \mathcal{S} \longrightarrow \mathfrak{g}_B, \quad u \wedge v \longmapsto K_{u,v}. \quad (4.3)$$

Theorem 4.2 (Bivector realization). *The map Φ is a linear equivalence. Consequently*

$$\dim_{\mathbb{R}} \mathfrak{g}_B = \dim_{\mathbb{R}} \Lambda^2 \mathcal{S} = 6. \quad (4.4)$$

Moreover, $\mathfrak{g}_B$ is closed under the commutator

$$[A, C] := AC - CA, \quad (4.5)$$

so it is a six-dimensional real Lie algebra.

Proof. Skewness of $K_{u,v}$ follows by expanding (4.2) and using the symmetry of B . Closure of $\mathfrak{g}_B$ under the commutator follows by applying the skewness identity twice:

$$B(ACx, y) = -B(Cx, Ay) = B(x, CAy),$$

and similarly with A and C interchanged.

It remains to prove that Φ is bijective. Let

$$\tau = (1, 0), \quad b_i = (0, \varepsilon_i), \quad i = 0, 1, 2,$$

where $\varepsilon_0, \varepsilon_1, \varepsilon_2$ is the fixed Euclidean basis of V . For $A \in \mathfrak{g}_B$, introduce the six coefficients

$$\begin{aligned} c_0 &= (A\tau)_V^0, & c_1 &= (A\tau)_V^1, & c_2 &= (A\tau)_V^2, \\ c_3 &= \langle (Ab_0)_V, \varepsilon_1 \rangle, & c_4 &= \langle (Ab_0)_V, \varepsilon_2 \rangle, & c_5 &= \langle (Ab_1)_V, \varepsilon_2 \rangle. \end{aligned} \quad (4.6a)$$

The B -skew condition determines every other matrix coefficient from these six: it forces $(A\tau)_0 = 0$, determines the scalar component of $A(0, u)$ by $\langle (A\tau)_V, u \rangle$, and makes the spatial block antisymmetric. Hence if all six numbers in (4.6a) vanish, then $A\tau = 0$ and $Ab_i = 0$ for $i = 0, 1, 2$. Since τ, b_0, b_1, b_2 span $\mathcal{S}$, one has $A = 0$. Thus the six coordinates are rigid.

Conversely, for $c = (c_0, \dots, c_5) \in \mathbb{R}^6$, define the explicit bivector

$$\begin{aligned} z(c) := & -c_0 \tau \wedge b_0 - c_1 \tau \wedge b_1 - c_2 \tau \wedge b_2 \\ & + c_3 b_0 \wedge b_1 + c_4 b_0 \wedge b_2 + c_5 b_1 \wedge b_2. \end{aligned} \quad (4.6b)$$

A direct evaluation of $K_{u,v}$ on the four basis vectors shows that the six coordinates of $\Phi(z(c))$ are exactly $c_0, \dots, c_5$. For any $A \in \mathfrak{g}_B$, taking c to be the six coordinates of A therefore gives an endomorphism $\Phi(z(c))$ with the same six coordinates as A ; rigidity yields $\Phi(z(c)) = A$. Thus Φ is surjective.

For injectivity, the map $z : \mathbb{R}^6 \rightarrow \Lambda^2 \mathcal{S}$ in (4.6b) is injective because the six coordinates of $\Phi(z(c))$ recover c . Since

$$\dim \mathbb{R}^6 = 6 = \dim \Lambda^2 \mathcal{S},$$

this injective linear map is also surjective. If $\Phi(z) = \Phi(z')$, write $z = z(c)$ and $z' = z(c')$; applying the six coordinate functionals gives $c = c'$, hence $z = z'$. Therefore Φ is injective as well, and so it is a linear equivalence. The dimension statement follows. $\square$

Boundary statement 4.3 (The first open infinitesimal edge). The statement

$$\mathrm{Lie}(G_{\mathrm{Spin}}) \simeq \mathfrak{g}_B \tag{4.7}$$

is not part of the present construction. The intrinsic G_{Spin} is currently certified as a topological group, not as a smooth Lie group. Accordingly, the differential

$$d\rho_e : \mathrm{Lie}(G_{\mathrm{Spin}}) \longrightarrow \mathfrak{g}_B \tag{4.8}$$

has not been constructed. These are theorem-level gaps between the group-level double cover and the already available Lie algebra $\mathfrak{g}_B$.

5 From local model to global base

The next step is deliberately not phrased as a consequence of the Spin group. The reason is structural: a fibre transition tells us how to compare fibres over a point already known to lie in two patches. It does not say which points of two local copies represent the same point of a global base.

5.1 Base-gluing data

Let I be an index set. A local piece is an open subset $D_i \subset \mathcal{S}$.

Definition 5.1 (Base-gluing datum). A base-gluing datum B consists of:

1. open sets $D_i \subset \mathcal{S}$;
2. open incidence domains $W_{ij} \subset D_i$;
3. continuous maps $\phi_{ij} : \mathcal{S} \rightarrow \mathcal{S}$, used on W_{ij} ;

such that, on the indicated domains,

$$\begin{aligned} W_{ii} &= D_i, & \phi_{ii}(x) &= x, \\ \phi_{ij}(W_{ij}) &\subset W_{ji}, & \phi_{ji}(\phi_{ij}(x)) &= x, \end{aligned} \tag{5.1}$$

and whenever $x \in W_{ij}$ and $\phi_{ij}(x) \in W_{jk}$,

$$x \in W_{ik}, \quad \phi_{jk}(\phi_{ij}(x)) = \phi_{ik}(x). \tag{5.2}$$

This is an explicit *sufficient* base-gluing primitive. We do not claim that every field in Definition 5.1 is separately necessary, or that this is the unique minimal encoding of incidence information.

Form the tagged disjoint union

$$X_0 := \coprod_{i \in I} D_i \tag{5.3}$$

and define

$$(i, x) \sim (j, y) \iff x \in W_{ij} \text{ and } \phi_{ij}(x) = y. \tag{5.4}$$

The axioms above make $\sim$ an equivalence relation. Define

$$M_B := X_0 / \sim \tag{5.5}$$

with the quotient topology.

Proposition 5.2 (Topological reconstruction). *The quotient map $X_0 \rightarrow M_B$ is open. Each local piece D_i embeds homeomorphically onto an open subset of M_B , and these images cover M_B . Thus M_B carries a charted-space structure modelled on $\mathcal{S}$.*

Proof. The saturation of an open subset of X_0 is open because each incidence set W_{ij} is open and ϕ_{ij} is a homeomorphism between W_{ij} and W_{ji} , with inverse ϕ_{ji} . Hence the quotient map is open. Equality of two points from the same tagged piece forces equality of their local coordinates by $\phi_{ii} = \text{id}$, so the map from each D_i is injective. Openness and continuity then give an open embedding. Surjectivity of the quotient map proves that the chart images cover. $\square$

Hausdorffness is not a formal consequence of Definition 5.1; the usual line-with-two-origins phenomenon already shows why a separation condition is needed.

Definition 5.3 (Closed gluing graph). The gluing graph of B is closed when

$$\Gamma_B := \{(p, q) \in X_0 \times X_0 : p \sim q\} \quad (5.6)$$

is closed.

Proposition 5.4 (Hausdorffness and second countability). *If Γ_B is closed, then M_B is Hausdorff. If, in addition, I is countable, then M_B is second countable.*

Proof. Let $q : X_0 \rightarrow M_B$ be the quotient map. By Proposition 5.2, q is continuous, open, and surjective. Consequently

$$q \times q : X_0 \times X_0 \longrightarrow M_B \times M_B$$

is again an open quotient map. The preimage of the diagonal is exactly the gluing relation:

$$(q \times q)^{-1}(\Delta_{M_B}) = \{(p, q) : p \sim q\} = \Gamma_B.$$

If Γ_B is closed, the quotient property therefore implies that Δ_{M_B} is closed. This is equivalent to Hausdorffness.

For second countability, each D_i is an open subspace of the finite-dimensional real vector space $\mathcal{S}$, hence is second countable. If I is countable, the tagged disjoint union $X_0 = \coprod_i D_i$ is second countable. An open quotient of a second-countable space is second countable: the images under q of a countable basis form a countable basis of M_B . Hence M_B is second countable. $\square$

5.2 Why the global base is additional

The need for incidence data is itself a theorem, not merely a warning.

Theorem 5.5 (Local pieces do not determine the base). *There exist two base-gluing data over the same two indexed local domains such that the local pieces agree exactly but their quotient bases are not homeomorphic.*

Proof. Take a nonempty connected open local domain $D \subset \mathcal{S}$ and two copies indexed by $\{\text{false}, \text{true}\}$. In the first gluing, make every pair of pieces fully incident and use the identity identification. In the second, take $W_{ij} = \emptyset$ for $i \neq j$ and again use the identity where defined. The first quotient is connected; the second is the disjoint union of two nonempty open components and is not connected. Hence no homeomorphism can exist between them. $\square$

Corollary 5.6. *No datum that only prescribes a fibre transition law over already specified overlaps can, by itself, determine the missing cross-piece incidence of the base.*

5.3 The smooth gate

For the reconstructed atlas, the coordinate change from piece i to piece j is precisely ϕ_{ij} on W_{ij} .

Definition 5.7 (Smooth gluing). A base-gluing datum is smoothly compatible when

$$\phi_{ij}|_{W_{ij}}$$

is C^∞ for every i, j .

Theorem 5.8 (Conditional smooth four-manifold). *Suppose:*

1. B is a base-gluing datum over $\mathcal{S}$;

2. its gluing graph is closed;
3. the index set I is countable;
4. B satisfies Definition 5.7.

Then M_B is Hausdorff, second countable, and a C^∞ four-manifold modelled on $\mathcal{S}$.

Proof. The Hausdorff and second-countable assertions are exactly Proposition 5.4. It remains to identify the smooth structure of the actual quotient atlas.

For each inhabited piece D_i , the open embedding of Proposition 5.2 yields a chart whose target is D_i . On an overlap of the i - and j -charts, the source of the coordinate change is not merely contained in the primitive incidence set: it is exactly W_{ij} . If $y \in W_{ij}$, the quotient relation identifies (i, y) with $(j, \phi_{ij}(y))$, and therefore the coordinate change is literally

$$y \mapsto \phi_{ij}(y).$$

Thus every transition map of the reconstructed atlas is a restriction of one of the functions already required to be C^∞ by Definition 5.7. The atlas consequently takes values in the C^∞ structure groupoid, so M_B is a smooth manifold. Finally, the model space is $\mathcal{S} = \mathbb{R} \oplus \mathbb{R}^3$, hence $\dim_{\mathbb{R}} \mathcal{S} = 4$.

Notice that the argument proves sufficiency of the smooth-gluing hypothesis for this reconstructed atlas. It does not claim the converse or necessity of every primitive smoothness field. $\square$

Control 5.9 (Smoothness is not automatic). There are base-gluing data satisfying the topological gluing axioms whose transition is a homeomorphism but not differentiable at the origin. An explicit control is

$$H(t, x)_0 = t, \quad H(t, x)_k = x_k + |t| \quad (k = 1, 2, 3),$$

with inverse

$$H^{-1}(t, x)_0 = t, \quad H^{-1}(t, x)_k = x_k - |t|.$$

Both maps are continuous, so the topological reconstruction applies. But along the line $t \mapsto (t, 0)$, every shifted spatial component contains $|t|$, which is not differentiable at $t = 0$. Hence the smooth-gluing condition fails.

Remark 5.10 (Sufficiency, not necessity). For the specific quotient-atlas packaging used here, smooth gluing is proved sufficient. A converse is not asserted. In particular, quotient representative selection may cause the reconstructed atlas to omit some index-labelled chart even when the corresponding primitive ϕ_{ij} is non-smooth.

6 Internal Spin data and the genuine tangent cocycle

Fix from now on a smooth base gluing B satisfying the hypotheses needed to obtain $M = M_B$.

6.1 Genuine tangent transitions

Because the actual atlas transition is ϕ_{ij} , the tangent bundle transition is its derivative.

Proposition 6.1 (Tangent transition law). *For $y \in W_{ij}$, the genuine tangent coordinate change of the reconstructed manifold is*

$$T_{ij}(y) := D\phi_{ij}(y) : \mathcal{S} \longrightarrow \mathcal{S}. \quad (6.1)$$

It is a linear isomorphism, with inverse

$$D\phi_{ji}(\phi_{ij}(y)). \quad (6.2)$$

The transitions satisfy the derivative cocycle law by the chain rule.

The invertibility in (6.2) is derived from $\phi_{ji} \circ \phi_{ij} = \text{id}$ on the open incidence domain rather than inserted as a separate hypothesis.

Remark 6.2 (A type-level shortcut is not a solder). In the formal implementation, tangent spaces of a modelled manifold are represented using the model vector space, so a type-level identification with $\mathcal{S}$ is available. This does not identify transition systems and plays no role in the geometric coupling below. The invariant datum is (6.1).

6.2 Native Spin transitions

Let $\mathcal{U} = \{U_i\}_{i \in I}$ be the cover of M by the chart images. A native Spin transition system consists of continuous maps

$$\tilde{g}_{ij} : U_i \cap U_j \longrightarrow G_{\text{Spin}} \quad (6.3)$$

satisfying the Čech cocycle law

$$\tilde{g}_{ij}(x)\tilde{g}_{jk}(x) = \tilde{g}_{ik}(x) \quad (6.4)$$

on triple overlaps. Projection gives the internal Lorentz cocycle

$$g_{ij} := \rho \circ \tilde{g}_{ij} : U_i \cap U_j \longrightarrow G_{\text{Lor}}. \quad (6.5)$$

This is a fibre transition system on the same cover, but it is not the tangent transition system by definition.

Theorem 6.3 (Raw transition identification is not forced). *There exist a smooth two-chart base gluing B and a coherent native Spin transition system $\tilde{g}$ such that*

$$g_{01}(x) = \text{id}_{\mathcal{S}} \quad \text{for every } x, \quad (6.6)$$

while

$$T_{01}(y) = 2 \text{id}_{\mathcal{S}} \quad \text{for every } y. \quad (6.7)$$

Hence the base and native Spin data do not force $T_{ij} = g_{ij}$.

Proof. Take two copies of the whole local model with

$$\phi_{01}(y) = 2y, \quad \phi_{10}(y) = \tfrac{1}{2}y.$$

All incidence domains are the whole model, and the gluing is smooth. Thus $D\phi_{01} = 2 \text{id}$. Independently, choose the pointwise trivial native Spin cocycle $\tilde{g}_{ij} = 1$; its projection is the identity. Evaluating at $\mathbf{1}_{\mathcal{S}}$ distinguishes the two transformations. $\square$

Remark 6.4. The conclusion is one of non-derivability, not non-existence. The same rescaling example admits a compensating regular solder. Equality of raw transition matrices is therefore not the correct invariant criterion for coupling the two bundles.

7 The regular solder gate

The missing object is a smoothly varying identification between the internal Lorentz fibres and tangent fibres that intertwines their transition laws.

7.1 Weak fibrewise soldering

At the weakest level one may choose, for each chart and each base point, a linear equivalence

$$e_i(x) : \mathcal{S} \xrightarrow{\sim} T_x M \quad (7.1)$$

such that on $U_i \cap U_j$,

$$e_j(x)v = e_i(x)(g_{ij}(x)v). \quad (7.2)$$

This algebraic condition is sufficient to transport the intrinsic form B to each tangent fibre, but it imposes no regularity on the field $x \mapsto e_i(x)$.

Control 7.1 (Weak soldering is insufficient). There exist fibrewise solder data obeying (7.2) whose chart representative is discontinuous. Consequently fibrewise linear equivalence alone does not produce a regular vector-bundle comparison or a smooth tangent metric.

Explicit control. Take the one-chart flat base with identity gluing and the pointwise trivial Spin cocycle. In this case (7.2) imposes no nontrivial overlap constraint. Let x_0 be the chart point represented by the zero vector and define the frame field

$$e(x) = \begin{cases} 2 \text{ id}_{\mathcal{S}}, & x = x_0, \\ \text{id}_{\mathcal{S}}, & x \neq x_0. \end{cases} \quad (7.2a)$$

Every fibre map is a linear equivalence, so this is a valid weak solder. Its local representative, however, jumps from id to 2id at x_0 , and is therefore discontinuous. The defect is also visible in the transported quadratic form: evaluated on the intrinsic unit, the coefficient is 1 away from x_0 , whereas at x_0 the inverse frame contributes a factor $1/2$, giving $1/4$. Thus even the induced fibrewise metric coefficient is discontinuous. No regular solder can induce this weak frame field. $\square$

7.2 Regular local form

The regular datum is stated in the actual chart coordinates. Let

$$A_i(y) \in \text{GL}(\mathcal{S})$$

be a continuous linear automorphism for $y \in D_i$.

Definition 7.2 (Regular solder). A regular solder for $(B, \tilde{g})$ is a family

$$A_i : D_i \longrightarrow \text{GL}(\mathcal{S}) \quad (7.3)$$

such that:

1. the map $y \mapsto A_i(y)$, viewed in the finite-dimensional space of continuous linear maps, is C^∞ on D_i ;
2. on $y \in W_{ij}$,

$$A_j(\phi_{ij}(y)) = T_{ij}(y) \circ A_i(y) \circ g_{ij}(x_y), \quad (7.4)$$

where $x_y \in M$ is the quotient point represented by (i, y) .

Equation (7.4) fixes the orientation of the intertwining square. The tangent map T_{ij} converts i -chart tangent components to j -chart tangent components, while the Čech convention $e_j = e_i \circ g_{ij}$ converts internal j -components to internal i -components. The equation is therefore forced by the two pre-existing conventions rather than selected by analogy.

The inverse family $A_i(y)^{-1}$ is automatically smooth; no separate inverse-regularity field is needed.

Theorem 7.3 (Regular bundle-equivalence certificate). *A regular solder determines a fibrewise linear equivalence between the internal Lorentz bundle defined by g_{ij} and the genuine tangent bundle TM , covering id_M , with:*

1. *continuous total-space map and continuous inverse;*
2. *linear equivalence on each fibre;*
3. *C^∞ local representatives A_i ;*
4. *C^∞ local representatives A_i^{-1} for the inverse;*
5. *exact intertwining of the internal and tangent transition systems.*

Conversely, a bundle-equivalence certificate with these properties yields a regular solder.

Proof. Write the internal Lorentz bundle in the local trivializations determined by the cocycle g_{ij} , and write TM in the tangent trivializations of the reconstructed atlas. On the i -th chart define the local total-space map

$$(y, v) \mapsto (y, A_i(y)v). \quad (7.4a)$$

If $y \in W_{ij}$, changing the source trivialization by g_{ij} and the target trivialization by T_{ij} gives, by (7.4), the same point as first applying (7.4a) in chart i . Hence the local maps agree on equivalent representatives and descend to a globally defined map of total spaces covering id_M . Each fibre map is a linear equivalence because each $A_i(y)$ is.

The local representative of the descended map in the i -th source and target trivializations is exactly A_i . Since A_i is smooth, it is continuous, and the standard local-trivialization criterion gives continuity of the total-space map. Pointwise inversion on $\text{GL}(\mathcal{S})$ is smooth; hence $y \mapsto A_i(y)^{-1}$ is smooth. The same argument applied to these inverse representatives yields a continuous inverse total-space map. This proves the five stated properties.

Conversely, suppose such a bundle-equivalence certificate is given. Read its local representative in the chart trivializations and call it $A_i(y)$. Fibrewise linear equivalence gives $A_i(y) \in \text{GL}(\mathcal{S})$, the certificate gives C^∞ -regularity, and equality of the globally defined map when read in the two overlapping trivializations is precisely the intertwining identity (7.4). Thus the local representatives form a regular solder. The two constructions are inverse at the level asserted in the theorem. $\square$

Remark 7.4 (Exact regularity level). We establish a bundle equivalence with continuous total-space maps in both directions, fibrewise linear equivalences, and smooth local representatives for the map and its inverse. No stronger smooth total-space equivalence is asserted here. In particular, the limitation concerns the level at which the global equivalence is packaged, not the smoothness of the projected Lorentz transition representatives, which is forced in chart coordinates by a regular solder.

Indeed, (7.4) can be solved for g_{ij} :

$$g_{ij}(x_y) = A_i(y)^{-1} \circ T_{ij}(y)^{-1} \circ A_j(\phi_{ij}(y)), \quad (7.5)$$

up to the convention used in (7.4). Hence regular soldering forces the projected internal Lorentz transition to possess smooth local representatives in chart coordinates even though the underlying native Spin cocycle is only assumed continuous.

7.3 Gauge freedom of the solder

For fixed B and fixed native Spin data, a solder is not canonical. The regular tangent-gauge group acts on the space of regular solders.

Theorem 7.5 (Solder torsor). *Whenever the regular-solder solution space for fixed $(B, \tilde{g})$ is nonempty, the regular tangent-gauge action on it is free and transitive.*

Proof. Let $A = \{A_i\}$ and $A' = \{A'_i\}$ be two regular solders for the same base and the same native Spin datum. Define

$$a_i(y) := A'_i(y) A_i(y)^{-1}. \quad (7.6)$$

The field a_i is smooth because multiplication and inversion in the finite-dimensional general linear group are smooth. Applying the solder law once to A and once to A' , and cancelling the common internal transition, gives

$$a_j(\phi_{ij}(y)) T_{ij}(y) = T_{ij}(y) a_i(y), \quad (7.7)$$

which is exactly the regular tangent-gauge compatibility law. By construction,

$$(a \cdot A)_i(y) = a_i(y) A_i(y) = A'_i(y),$$

so the action is transitive.

For freeness, if $a \cdot A = a' \cdot A$, then $a_i(y) A_i(y) = a'_i(y) A_i(y)$ at every point. Right cancellation by the invertible map $A_i(y)$ gives $a_i(y) = a'_i(y)$ for all i, y , hence $a = a'$. Pointwise composition and inversion preserve (7.7), so these fields form a group and the action is free and transitive. $\square$

Thus existence is a genuine gate, while uniqueness is neither expected nor claimed.

8 The induced tangent Lorentz geometry

Let $A = \{A_i\}$ be a regular solder.

8.1 Metric coefficients

In chart i , define

$$g_i(y)(u, v) := B(A_i(y)^{-1}u, A_i(y)^{-1}v). \quad (8.1)$$

Theorem 8.1 (Smooth tangent Lorentz metric). *The fields g_i satisfy:*

1. *for fixed $u, v \in \mathcal{S}$, the function $y \mapsto g_i(y)(u, v)$ is C^∞ on D_i ;*
2. *$g_i(y)$ is symmetric and nondegenerate;*
3. *on every incidence domain,*

$$g_j(\phi_{ij}(y))(T_{ij}(y)u, T_{ij}(y)v) = g_i(y)(u, v); \quad (8.2)$$

4. *in the $A_i(y)$ -frame, the form has signature $(1, 3)$.*

Hence the g_i satisfy the smooth tensor transformation law and therefore define a global Lorentzian $(0, 2)$ -tensor g on TM , in the standard semi-Riemannian sense [10, 9]. This conclusion uses the smooth local coefficient fields and their exact tensor transformation law; no stronger global metric packaging is asserted.

Proof. Smoothness follows from smoothness of A_i^{-1} and bilinearity of B . Symmetry and non-degeneracy are inherited from B through the linear equivalence $A_i(y)$. For the transformation law, use (7.4) to express the inverse j -frame after application of T_{ij} in terms of the inverse i -frame and the projected internal Lorentz transition. Since $g_{ij}(x) \in G_{\text{Lor}}$ preserves B , these Lorentz factors cancel and yield (8.2). Finally,

$$g_i(y)(A_i(y)a, A_i(y)b) = B(a, b),$$

so Proposition 2.3 transfers the signature. $\square$

The important logical point is that the metric is *derived after* regular soldering. It is not a field of the base-gluing datum and is not inserted into the primitive local data.

8.2 Orientation reduction

Let

$$F_{ij}(y) := A_j(\phi_{ij}(y))^{-1} \circ T_{ij}(y) \circ A_i(y). \quad (8.3)$$

By the solder law this is the appropriate projected internal Lorentz transition (with the inverse-index convention induced by (7.4)). Therefore

$$\det F_{ij}(y) = 1. \quad (8.4)$$

Proposition 8.2 (Project-native orientation reduction). *The regular solder frames form a coherently oriented family: their frame changes have determinant 1. Equivalently, the determinant cocycle of the genuine tangent transitions is a continuous coboundary determined by the solder-frame determinants.*

8.3 Future-cone reduction

Define the intrinsic interior future cone exactly by

$$\mathcal{C}_{\text{int}}^+ := \{x \in \mathcal{S} : x \in C_{\mathcal{S}} \text{ and } N(x) > 0\}.$$

This uses only the already constructed square cone and intrinsic quadratic form. In chart i , define

$$\mathcal{C}_i^+(y) := A_i(y)\mathcal{C}_{\text{int}}^+. \quad (8.5)$$

Proposition 8.3 (Project-native future-cone reduction). *For $y \in W_{ij}$,*

$$u \in \mathcal{C}_i^+(y) \iff T_{ij}(y)u \in \mathcal{C}_j^+(\phi_{ij}(y)). \quad (8.6)$$

Thus the regular solder determines a coherent chart-independent field of future cones on TM .

The local vector field

$$\tau_i(y) := A_i(y)\mathbf{1}_{\mathcal{S}} \quad (8.7)$$

is smooth, future-directed, and unit timelike. It need not glue to a single global field. More precisely, the distinguished fields τ_i glue across an overlap exactly when the corresponding projected Lorentz transition fixes $\mathbf{1}_{\mathcal{S}}$.

Boundary statement 8.4 (Time-orientability terminology). The conclusion formally established here is the coherent future-cone reduction (8.6). No comparison theorem in the present development identifies this project-native structure with every standard equivalent definition of time orientability. Likewise, existence of an arbitrary global smooth future-directed timelike section would require an additional global section theorem in the present bundle packaging.

9 Native Čech lifting and global reconvergence

We now compare the bottom-up construction with the obstruction-theoretic language that ordinarily appears in Spin geometry.

9.1 Fixed-cover lifting obstruction

Let

$$g_{ij} : U_i \cap U_j \rightarrow G_{\text{Lor}}$$

be a visible Lorentz cocycle. A family of local Spin lifts consists of continuous maps

$$h_{ij} : U_i \cap U_j \rightarrow G_{\text{Spin}}, \quad \rho(h_{ij}) = g_{ij}. \quad (9.1)$$

Local sections of ρ make such lifts available locally. On triple overlaps, define the defect

$$\epsilon_{ijk} := h_{ij} h_{jk} h_{ik}^{-1}. \quad (9.2)$$

Because $\rho(\epsilon_{ijk}) = 1$,

$$\epsilon_{ijk} \in \ker \rho = \{\pm 1\}. \quad (9.3)$$

This defines a fixed-cover kernel-valued obstruction class. The independence from the chosen lift family is elementary at this level. If h'_{ij} is another lift of the same visible Lorentz cocycle, then there are kernel-valued functions $\alpha_{ij} : U_i \cap U_j \rightarrow \{\pm 1\}$ with

$$h'_{ij} = \alpha_{ij} h_{ij}.$$

Because the kernel is central,

$$\begin{aligned} \epsilon'_{ijk} &= h'_{ij} h'_{jk} (h'_{ik})^{-1} \\ &= \alpha_{ij} \alpha_{jk} \alpha_{ik}^{-1} \epsilon_{ijk}. \end{aligned} \quad (9.3a)$$

Thus two representatives differ by the fixed-cover kernel-valued coboundary $\delta\alpha$, and determine the same project-native class. A coherent native Spin cocycle gives $\epsilon_{ijk} = 1$ identically, so its class is trivial.

This is the familiar mechanism underlying the usual obstruction theory for Spin structures [11, 8, 6]; however, the present class is kept at its exact project-native fixed-cover level.

Boundary statement 9.1 (No Stiefel–Whitney identification). No theorem here identifies the determinant obstruction with $w_1(TM)$, or the fixed-cover kernel-valued lifting obstruction with $w_2(TM)$. No refinement limit or comparison with global $H^1(M; \mathbb{Z}/2)$ or $H^2(M; \mathbb{Z}/2)$ is used. Consequently the statement $w_1(TM) = w_2(TM) = 0$ is not a theorem of the present paper.

9.2 Reconvergence under the solder gate

The regular solder turns the projected internal Lorentz cocycle into the actual tangent Lorentz cocycle read in the solder frames.

Theorem 9.2 (Solder-induced cocycle identification). *For every regular solder and every overlap, the Lorentz transformation obtained by comparing tangent components in the two solder frames is exactly the projected native transition g_{ij} .*

This is a rearrangement of the intertwining equation (7.4), but it is the central bridge: it allows the internal double-cover information to be read as information about the tangent bundle.

Theorem 9.3 (Global reconvergence). *Assume B is smoothly glued and a regular solder exists for a native Spin cocycle $\tilde{g}$. Then:*

1. there are continuous nowhere-zero functions $a_i : D_i \rightarrow \mathbb{R}$ such that

$$\det T_{ij}(y) a_i(y) = a_j(\phi_{ij}(y)); \quad (9.4)$$

2. the tangent Lorentz cocycle in solder frames equals $\rho(\tilde{g}_{ij})$;

3. every local lift family of that projected cocycle has trivial project-native fixed-cover Spin-lifting obstruction;

4. the project-native future-cone reduction of Proposition 8.3 exists.

Proof. For (1), take the determinant of the regular-solder intertwining law. The internal Lorentz factor has determinant one, so the determinant of the tangent transition is the ratio of the solder-frame determinants, giving (9.4). Statement (2) is theorem 9.2. For (3), the native Spin cocycle itself is a coherent lift of its projection; the triple-overlap defect is therefore trivial. Independence of the obstruction from the chosen local lift family transfers triviality to every lift family. Statement (4) is Proposition 8.3. $\square$

Two useful contrapositives are immediate.

Corollary 9.4 (Orientation gate). *If no continuous nowhere-zero 0-cochain a_i can satisfy (9.4), then no regular solder exists for any native Spin datum over that base gluing.*

Corollary 9.5 (Spin-lifting gate). *If a visible Lorentz cocycle has nontrivial project-native lifting obstruction, then it admits no coherent native Spin transition datum projecting to it. In particular it cannot arise as the tangent Lorentz cocycle of a regularly soldered native Spin datum.*

The converse of theorem 9.3 is not proved. Vanishing of these two necessary obstructions is therefore not asserted to be sufficient for a regular solder.

10 Controls and residual global Spin freedom

The previous theorems are accompanied by controls designed to distinguish logical necessity from convenient examples.

10.1 One-chart positive control

On a one-chart base there are no nontrivial overlap constraints. The base is smooth, a regular solder exists, and there is no artificial sign family generated by the cover. This verifies that the closure machinery does not manufacture an obstruction where none is present.

10.2 Orientation-reversing negative control

There is a smooth Möbius-type base gluing whose cross-overlap has two relevant components. On the wrap-around component the genuine tangent transition has determinant -1 , whereas on the ordinary overlap component it has determinant $+1$.

Control 10.1 (Orientation reversal excludes regular solder). For the orientation-reversing control, no regular solder exists for *any* native Spin transition datum.

Reason. Every projected native Spin transition lies in G_{Lor} and hence has determinant $+1$. If a regular solder existed, the determinant of each local solder frame would be a continuous nowhere-zero function on a connected chart domain and therefore would have constant sign there. The solder determinant law on the wrap-around component, where the tangent determinant is -1 , forces opposite signs in the two charts; the same law on the ordinary component, where the tangent determinant is $+1$, forces the same signs. These requirements are incompatible. This is the concrete mechanism behind the orientation gate Corollary 9.4; the presence of both overlap components is essential. $\square$

Native Spin transition data nevertheless exist over this same gluing. Thus existence of Spin-valued fibre data is strictly weaker than existence of a Lorentz solder coupling those data to TM .

10.3 Combinatorial-label negative control

Consider also a deliberately abstract Spin-foam-style label structure. For arbitrary such combinatorial labels, the same label input is compatible with the one-chart positive control, where a regular solder exists, and with the Möbius-type control, where no regular solder exists for any native Spin datum. Therefore these combinatorial labels do not determine a regular solder. This is a statement about this abstract control structure, not about any published spin-foam model or about spin foams in general.

10.4 Fixed-cover kernel freedom

Because $\ker \rho = \{\pm 1\}$, a native lift can retain discrete global freedom even when the projected Lorentz transition system is fixed.

Proposition 10.2 (One-loop sign freedom). *On the fixed one-loop control cover there are two native Spin transition systems that:*

1. *have the same projected Lorentz transition system;*
2. *both admit regular solders;*
3. *are not equivalent under the project's kernel-valued fixed-cover gauge relation.*

Proposition 10.3 (Finite fixed-cover multiplicity). *For N independent loop labels, the construction gives an injection of $\{\pm 1\}^N$ into the fixed-cover gauge classes. In particular, for $N = 2$ there are four pairwise distinguished classes, all compatible with regular soldering in the control.*

Proof of Propositions 10.2 and 10.3. For each loop n , the cross-overlap of its two patches has two connected components: the wrap-around component and the ordinary component. Given a sign assignment $\sigma \in \{\pm 1\}^N$, define a kernel-valued Čech 1-cocycle z^σ which equals σ_n on the wrap-around component of loop n and equals 1 on the ordinary component and on all unaffected overlaps. Since every value of z^σ lies in $\ker \rho = \{\pm 1\}$, twisting the trivial Spin cocycle by z^σ leaves the projected Lorentz cocycle unchanged. Consequently the same regular solder for the periodic control works for every sign assignment.

It remains to prove that distinct sign assignments are not related by a kernel-valued fixed-cover gauge. Suppose $\sigma_n \neq \tau_n$ for some loop n , and assume that patchwise kernel-valued gauge functions λ_- and λ_+ relate the corresponding twisted cocycles. The kernel is discrete and each patch is connected, so continuity forces λ_- and λ_+ to be constant on their patches. Evaluating the gauge law at a point of the ordinary overlap, where both sign cocycles equal 1, gives

$$\lambda_- \lambda_+^{-1} = 1.$$

Evaluating the same law at a point of the wrap-around component gives, using centrality of the kernel,

$$\tau_n = \lambda_- \sigma_n \lambda_+^{-1} = \sigma_n,$$

contradicting $\sigma_n \neq \tau_n$. Hence different sign assignments give different fixed-cover gauge classes. For one loop this yields exactly the two classes of Proposition 10.2; for N independent loops it gives the asserted injection, and for $N = 2$ the four sign assignments give four pairwise inequivalent classes. $\square$

Boundary statement 10.4. The preceding statements do not compute $H^1(M; \mathbb{Z}/2)$. They are fixed-cover statements; no refinement argument identifying them with global cohomology is part of the present construction.

This control is important conceptually: bottom-up Spin closure does not canonize the Spin lift.

11 A selected one-parameter transition deformation

The static closure structure is now complete enough to perform a first deformation smoke test. The deformation is placed at the strongest transition-level object that presently exists: the native Spin cocycle. There is no connection one-form, path-ordered transport, holonomy functional, curvature two-form, or field equation in the construction.

11.1 Selected intrinsic one-parameter subgroup

Choose the first Euclidean basis direction $\varepsilon_0 \in V$ and its Clifford image

$$\gamma_0 := \iota(\varepsilon_0), \quad \gamma_0^2 = 1.$$

The generator γ_0 is selected for the control; no theorem makes this direction canonical.

Definition 11.1 (Selected Spin path). For $\lambda \in \mathbb{R}$, define

$$G(\lambda) := \cosh\left(\frac{\lambda}{2}\right) + \sinh\left(\frac{\lambda}{2}\right) \gamma_0 \in \text{Cl}_3. \quad (11.1)$$

The hyperbolic identities and $\gamma_0^2 = 1$ give

$$G(\lambda)c(G(\lambda)) = 1,$$

hence $G(\lambda) \in G_{\text{Spin}}$. Moreover,

$$G(0) = 1, \quad G(\lambda + \mu) = G(\lambda)G(\mu), \quad G(-\lambda) = G(\lambda)^{-1}. \quad (11.2)$$

The parameterization is injective.

The projection $\rho(G(\lambda))$ is nontrivial for $\lambda \neq 0$. On the plane spanned by the intrinsic scalar direction and $(0, \varepsilon_0)$, its action is the standard boost block

$$\begin{pmatrix} \cosh \lambda & \sinh \lambda \\ \sinh \lambda & \cosh \lambda \end{pmatrix}, \quad (11.3)$$

and it acts trivially on the two orthogonal Euclidean directions. In particular,

$$\rho(G(\lambda))\mathbf{1}_{\mathcal{S}} = (\cosh \lambda, \sinh \lambda \varepsilon_0). \quad (11.4)$$

Boundary statement 11.2 (Selected control slice). The family (11.1) is a deliberately selected one-parameter subgroup generated by one chosen Clifford direction. It is neither proved unique nor canonical, and it does not exhaust the possible intrinsic Spin deformation directions.

11.2 The fixed periodic control

Fix the periodic base gluing used by the control, together with its fixed cover and two-component periodic overlap. We make no identification of this quotient with $S^1 \times \mathbb{R}^3$, and no fundamental-group theorem is required.

The native Spin transition cocycle is deformed using $G(\lambda)$. The base gluing and hence the tangent transitions are held fixed; the projected Lorentz transition is always obtained by applying ρ to the Spin transition. The regular solder is then the variable to be solved for.

Let

$$\mathcal{R}_A := \{\lambda \in \mathbb{R} : \text{a regular solder exists for the deformed cocycle at } \lambda\}. \quad (11.5)$$

Theorem 11.3 (Admissibility locus). *For the fixed periodic control,*

$$\mathcal{R}_A = \mathbb{R}. \quad (11.6)$$

Thus every finite parameter value admits an explicitly constructed regular solder. In particular admissibility is symmetric under $\lambda \mapsto -\lambda$.

Proof. Let $\chi : \mathbb{R} \rightarrow [0, 1]$ be a fixed C^∞ transition profile with $\chi(t) = 0$ for $t \leq 3$ and $\chi(t) = 1$ for $t \geq 4$. On the two patches of each periodic control loop define

$$\alpha_\lambda(p, y) = \begin{cases} 0, & p \text{ is the first patch,} \\ \lambda \chi(y_0), & p \text{ is the second patch.} \end{cases} \quad (11.6a)$$

Let $P(t) = \rho(G(t)) \in G_{\text{Lor}}$. By the one-parameter subgroup law, $P(s+t) = P(s)P(t)$ and $P(-t) = P(t)^{-1}$, and the closed form (11.3) makes $t \mapsto P(t)$ smooth. Define

$$A_p^{(\lambda)}(y) := P(\alpha_\lambda(p, y)). \quad (11.6b)$$

This is the identity on the first patch and smoothly interpolates on the second patch between the identity and $P(\lambda)$.

All tangent transitions of the frozen periodic control are the identity. The projected deformed Spin transition on an overlap is $P(\beta_{pq}^{(\lambda)}(x))$, where $\beta_{pq}^{(\lambda)}$ is $\pm\lambda$ on the wrap-around component and 0 on the ordinary component. By the defining values of χ , one verifies on each overlap component that

$$\alpha_\lambda(q, \phi_{pq}(y)) = \alpha_\lambda(p, y) + \beta_{pq}^{(\lambda)}(x_y). \quad (11.6c)$$

Therefore

$$\begin{aligned} A_q^{(\lambda)}(\phi_{pq}(y)) &= P(\alpha_\lambda(q, \phi_{pq}(y))) \\ &= P(\alpha_\lambda(p, y))P(\beta_{pq}^{(\lambda)}(x_y)) \\ &= A_p^{(\lambda)}(y) g_{pq}^{(\lambda)}(x_y), \end{aligned}$$

which is precisely the regular-solder law because the tangent transition is id. Smoothness and pointwise invertibility follow from those of P and χ . Hence a regular solder exists for every $\lambda \in \mathbb{R}$, proving $\mathcal{R}_A = \mathbb{R}$. $\square$

The result is not vacuous: the project also contains closure families for which not every parameter is admissible. Equation (11.6) is therefore a property of this particular control family, not a consequence of the definition of admissibility.

11.3 Full Spin Čech gauge orbit

Let $\tilde{g}^{(\lambda)}$ denote the native transition cocycle at parameter λ .

Theorem 11.4 (Gauge classification of the selected family). *For all $\lambda_1, \lambda_2 \in \mathbb{R}$,*

$$\tilde{g}^{(\lambda_1)} \sim_{\text{full Spin Čech}} \tilde{g}^{(\lambda_2)}. \quad (11.7)$$

The Spin-side gauge is continuous. After projection by ρ , its Lorentz representative is C^∞ on every chart domain, and the projected Lorentz cocycles also lie in one gauge orbit.

Proof. First compare an arbitrary λ with the neutral value. Using the same interpolation function α_λ as in (11.6a), define on patch p

$$u_p^{(\lambda)}(x) := G(-\alpha_\lambda(p, \text{coord}_p(x))). \quad (11.7a)$$

The map is continuous because $G(t)$ is continuous in the intrinsic Spin topology and the chart-coordinate profile is continuous. On an overlap, (11.6c) and the group law of G give

$$\tilde{g}_{pq}^{(\lambda)}(x) = u_p^{(\lambda)}(x) \tilde{g}_{pq}^{(0)}(x) (u_q^{(\lambda)}(x))^{-1}. \quad (11.7b)$$

Thus every parameter value is full-Spin-Čech gauge equivalent to zero. For two arbitrary values λ_1, λ_2 , compose the inverse gauge from λ_1 to zero with the gauge from zero to λ_2 .

Applying ρ to (11.7b) gives the corresponding Lorentz gauge relation. Its chartwise representative is $P(-\alpha_\lambda)$; by the explicit hyperbolic formula (11.3) and smoothness of α_λ , this projected gauge is C^∞ on every chart domain. No smooth structure on the intrinsic Spin group is used or claimed. $\square$

The narrower kernel-valued fixed-cover equivalence behaves differently. For $\lambda \neq 0$, the deformed datum is not merely a $\{\pm 1\}$ -lift relabeling of the neutral datum, because its Lorentz projection itself moves. There is no contradiction: the two equivalence relations answer different questions.

11.4 All regular solder solutions, not only one branch

For a parameter λ , write

$$\mathcal{E}_\lambda := \left\{ \text{regular solders for } (B_{\text{per}}, \tilde{g}^{(\lambda)}) \right\}. \quad (11.8)$$

Theorem 11.5 (Solution-space correspondence). *For every $\lambda_1, \lambda_2 \in \mathbb{R}$, there is an explicit bijection*

$$\mathcal{T}_{\lambda_1, \lambda_2} : \mathcal{E}_{\lambda_1} \xrightarrow{\sim} \mathcal{E}_{\lambda_2}, \quad (11.9)$$

obtained by precomposing the local solder comparison with the projected gauge relating the two native Spin cocycles. The inverse is $\mathcal{T}_{\lambda_2, \lambda_1}$.

Proof. Set

$$\delta_{12}(p, y) := \alpha_{\lambda_2}(p, y) - \alpha_{\lambda_1}(p, y), \quad Q_{12,p}(y) := P(\delta_{12}(p, y)). \quad (11.9a)$$

For a regular solder $A \in \mathcal{E}_{\lambda_1}$, define

$$(\mathcal{T}_{\lambda_1, \lambda_2} A)_p(y) := A_p(y) \circ Q_{12,p}(y). \quad (11.9b)$$

The fields $Q_{12,p}$ are smooth and invertible. Subtracting the two identities (11.6c) for λ_1 and λ_2 gives exactly the overlap equation needed to convert the λ_1 -solder law into the λ_2 -solder law. Thus (11.9b) is a regular solder at λ_2 .

Since

$$\delta_{21} = -\delta_{12} \quad \text{and} \quad P(-t) = P(t)^{-1},$$

transporting from λ_1 to λ_2 and then back composes with $Q_{12}Q_{21} = \text{id}$ pointwise. Hence $\mathcal{T}_{\lambda_2, \lambda_1}$ is the inverse of $\mathcal{T}_{\lambda_1, \lambda_2}$, proving the stated bijection. $\square$

Theorem 11.6 (Metric preservation). *Let $A \in \mathcal{E}_{\lambda_1}$, and let $A' = \mathcal{T}_{\lambda_1, \lambda_2}(A)$. Then for every $x \in M$ and $X, Y \in T_x M$,*

$$g_{A'}(X, Y) = g_A(X, Y). \quad (11.10)$$

Proof. In a chart, (11.9b) gives

$$A'_p = A_p \circ Q_{12,p}, \quad (A'_p)^{-1} = Q_{12,p}^{-1} \circ A_p^{-1}.$$

Therefore

$$\begin{aligned}
g_{A',p}(u, v) &= B\left(Q_{12,p}^{-1}A_p^{-1}u, Q_{12,p}^{-1}A_p^{-1}v\right) \\
&= B\left(A_p^{-1}u, A_p^{-1}v\right) \\
&= g_{A,p}(u, v),
\end{aligned}$$

because $Q_{12,p}(y) = P(\delta_{12}(p, y)) \in G_{\text{Lor}}$ preserves B . The equality holds in every chart and is compatible with the tangent transition law, hence it is the equality of the induced tangent metrics at every base point and on every pair of tangent vectors. $\square$

Corollary 11.7 (Scoped negative smoke-test conclusion). *On the fixed periodic control, the selected transition-cocycle parameter has no closure-selection power and does not change the induced tangent Lorentz metric under the certified correspondence of regular-solder solutions.*

Boundary statement 11.8 (What Corollary 11.7 does not say). The result does not classify arbitrary native Spin deformations, arbitrary covers, co-varying bases, or connection-level deformations. It does not show that “all transition deformations are pure gauge”, and it carries no curvature interpretation. The parameter λ is the coordinate of the selected Spin subgroup (11.1), not a curvature scalar.

12 Mathematical boundary and next structural edge

The construction reaches a sharp stopping point. It is useful to state it positively before listing what remains open.

From the sole local primitive $V = \mathbb{R}^3$ with its Euclidean form, one has constructed the Lorentz spin factor, the Euclidean Clifford envelope, the intrinsic Spin double cover, and the six-dimensional Lorentz-skew algebra. After adding explicit base-gluing data and sufficient topological and smooth conditions, one obtains a smooth four-manifold. After adding a regular solder, one obtains a smooth Lorentzian tangent metric, coherent orientation and future-cone reductions, and the exact identification of the tangent Lorentz cocycle with the projection of native Spin transition data.

What is *not* yet part of the mathematical object is equally precise.

1. The intrinsic topological group G_{Spin} has not been packaged and related to $\mathfrak{g}_B$ as a Lie group/Lie algebra pair:

$$\text{Lie}(G_{\text{Spin}}) \simeq \mathfrak{g}_B \quad \text{is open.} \quad (12.1)$$

2. The differential $d\rho_e$ of the native Spin cover has not been constructed.
3. No Spin or Lorentz connection one-form has been defined.
4. Consequently there is no covariant derivative, parallel transport, holonomy, curvature, torsion, action functional, or field equation in the present theory.
5. The fixed-cover obstruction classes have not been compared formally with $w_1(TM)$ and $w_2(TM)$.
6. The project-native future-cone reduction has not been identified by theorem with all standard formulations of time orientability.

The selected deformation result of section 11 gives a concrete reason not to skip these missing edges. A nontrivial curve of transition-level Spin data can move the projected Lorentz cocycle while remaining fully gauge absorbable on the tested control and leaving the tangent metric invariant. Thus, on this control, the first tested transition-level knob has no closure-selection power and does not change the induced tangent Lorentz metric under the certified solder correspondence. This does not prove that a connection-level theory will be nontrivial; it identifies the next mathematically meaningful place at which such a question can be posed.

13 Conclusion

The main structural lesson is a separation theorem for levels of geometric information.

At the local level, positive Euclidean three-space already contains enough algebraic structure to produce a four-dimensional Lorentz spin factor, its Clifford envelope, an intrinsic Spin double cover of an intrinsic Lorentz group, and a six-dimensional Lorentz-skew Lie algebra. This part is intrinsic to the local input.

At the global level, incidence is independent information. Local pieces, even when equipped with fibre transitions, do not determine which points of different pieces form a common base. Explicit base gluing is therefore a genuine additional primitive. Under a closed-graph separation condition, countability, and smooth compatibility, it reconstructs a smooth four-manifold.

At the tangent level, a second independent gap appears. The derivative cocycle $D\phi_{ij}$ of that manifold is not forced to equal the internal Lorentz cocycle. A regular solder is the additional gate that identifies them in the correct bundle-theoretic sense. Once supplied, it yields a smooth Lorentzian tangent metric and forces the project-native orientation, future-cone, and Spin-lifting compatibility conditions.

The resulting implication is therefore conditional and exact:

$$\begin{array}{c}
 (\mathbb{R}^3, \langle \cdot, \cdot \rangle) \implies (\mathcal{S}, N, \text{Cl}_3, G_{\text{Spin}} \rightarrow G_{\text{Lor}}, \mathfrak{g}_B) \\
 \text{local core + base gluing} \\
 \text{+closed gluing graph + countable index + smoothness} \implies M^4 \\
 M^4 + \text{native Spin cocycle + regular solder} \\
 \Downarrow \\
 (TM, g), \\
 \text{project-native orientation compatibility,} \\
 \text{project-native future-cone reduction,} \\
 \text{tangent Lorentz cocycle = projected native Spin cocycle,} \\
 \text{trivial project-native fixed-cover Spin-lifting obstruction.}
 \end{array} \tag{13.1}$$

No arrow in (13.1) hides a missing global existence theorem.

The first deformation smoke test is correspondingly modest but sharp. On the selected fixed periodic control, the chosen one-parameter Spin transition family is globally admissible and gauge absorbable for all finite parameters, and its entire regular-solder solution spaces are identified without changing the induced tangent Lorentz metric. This closes that specific control question and leaves the infinitesimal Spin-to-Lorentz bridge, and only thereafter connection-level geometry, as the next structural problem.

A Dependency and status table

Object or statement	Status	Dependency
$V = \mathbb{R}^3$, Euclidean form	primitive input	none
$\mathcal{S}, N, B$, signature (1, 3)	DERIVED	Euclidean input
Cl_3 , paravectors	DERIVED	q_3
$G_{\text{Spin}} \rightarrow G_{\text{Lor}}$, kernel $\{\pm 1\}$	DERIVED	Clifford core
$\mathfrak{g}_B \simeq \Lambda^2 \mathcal{S}$	DERIVED	B
Base incidence/gluing	ADDITIONAL DATUM	not determined by local pieces
Closed gluing graph	additional hypothesis	needed for certified Hausdorffness
Countable index set	additional hypothesis	needed for certified second countability
Smooth gluing	sufficient additional hypothesis	base transitions ϕ_{ij}
Smooth M^4	DERIVED	preceding global hypotheses
Tangent transitions $D\phi_{ij}$	DERIVED	smooth atlas
Native Spin cocycle	global datum	fixed cover of M
Regular solder	ADDITIONAL DATUM	not generally inhabited
Smooth Lorentz tangent metric	DERIVED	regular solder
Orientation reduction	DERIVED	regular solder
Future-cone reduction	DERIVED	regular solder
Project-native Spin obstruction trivial	DERIVED	regular solder + native cocycle
w_1, w_2 comparison	OPEN	characteristic-class comparison theorem
$\text{Lie}(G_{\text{Spin}}) \simeq \mathfrak{g}_B$	OPEN	smooth intrinsic Spin theory
$d\rho_e$	OPEN	preceding infinitesimal bridge
Connection, holonomy, curvature	OPEN	connection-level extension

B Formal source map

This appendix is a reproducibility map, not part of the mathematical dependency argument. The human-readable theorem statements above are tied to the current formal source by the following principal declarations.

Paper statement	Principal formal endpoint(s)
Spin factor and Lorentz form	<code>SpinCore.sJ_jordan</code> , <code>SpinCore.lorentz_quadratic_form</code>
Jordan–Clifford bridge	<code>SpinCore.clifford_jordan</code> , <code>SpinCore.adjoin_spinToCl_eq_top</code>
Intrinsic double cover	<code>SpinCore.spin_double_cover_bundled</code> , <code>SpinCore.spin_double_cover_topological_endpoint</code> , <code>SpinCore.spin_double_cover_local_section_endpoint</code>
Bivector/Lorentz-skew algebra	<code>SpinCore.bivectorEquivSkew</code> , <code>SpinCore.finrank_gB</code> , <code>SpinCore.gBLie_bracket</code>
Base non-derivability	<code>EmergentBase.base_not_determined_by_local_pieces</code>
Topological reconstruction	<code>EmergentBase.BaseGluingData.emergentManifoldCertificate</code>
Smooth manifold	<code>EmergentBase.BaseGluingData.isManifold_of_smoothGluing</code> , <code>EmergentBase.BaseGluingData.smoothEmergentManifoldCertificate</code>
Genuine tangent transition	<code>EmergentBase.BaseGluingData.tangentTransition_eq_derivative_baseTransition</code>
Tangent/internal independence	<code>SpinNative.tangent_spin_base_data_do_not_force_transition_identification</code>
Regular solder	<code>SpinNative.SmoothTangentSolderData</code> ; <code>SpinNative.smooth_solder_iff_regular_bundle_equivalence</code>
Smooth tangent metric	<code>SpinNative.SmoothTangentSolderData.smooth_tangent_lorentz_metric</code>
Orientation/future cone	<code>SpinNative.SmoothTangentSolderData.orientation_time_orientation_status</code>
Global reconvergence	<code>Task36.classical_reconvergence</code> , <code>Task36.spin_obstruction_nonzero_implies_no_solder</code>
Adversarial controls	<code>Task36.adversarial_global_topology_certificate</code> , <code>Task36.spin_foam_labels_do_not_determine_solder</code>
Selected deformation path	<code>Task37.Deformation.transportState_add</code> , <code>Task37.Deformation.projectedTransportState_apply</code>
Fixed-control admissibility	<code>Task37.Deformation.loopParaSolder (construction)</code> , <code>Task37.Deformation.controlAAdmissible_all</code> , <code>Task37.Deformation.regularRegion_loopTransportFamily</code>
Full Spin Čech gauge orbit	<code>Task37.Deformation.loopGaugeFun (construction)</code> , <code>Task37.Deformation.spinCocycle_gaugeEquiv</code> , <code>Task37.Deformation.projectedCocycle_gaugeEquiv</code>
Selected deformation smoke test	<code>Task37.Deformation.native_transport_knob_smoketest_certificate</code>
Whole solution-space transport	<code>Task37.Deformation.loopSolderEquiv (construction)</code> , <code>Task37.Deformation.loopSolderTransport_involutive</code> , <code>Task37.Deformation.loopSolderSolutionSpace_correspondence</code> , <code>Task37.Deformation.tangentMetric_loopSolderTransport</code>

The formal development is available at https://github.com/antaris82/The_Relational_Spin-Closure_Hypothesis [14]. The main development is written in Lean 4 and uses Mathlib [12, 13]. The mathematical claims of this paper should be read at the strength of the displayed hypotheses above; the formal artifact supplies machine-checked proofs of the corresponding registered source statements.

Acknowledgments

The editor thanks Thomas Stoer and Thomas Klimpel for sustained, critical, and constructive discussions. These exchanges repeatedly helped clarify assumptions, expose weak formulations, and sharpen the mathematical questions addressed in this work.

AI and formalization disclosure

ChatGPT (OpenAI), acting as AI Assistant, was used for mathematical exploration, source-level analysis, organization, consistency checking, and editorial drafting. Aristotle was used as Lean Agent for prover-assisted formalization and verification. The editor curated the scope and the presentation. Mathematical claims are asserted only at the strength stated in the paper and are tied, where applicable, to the referenced formal Lean endpoints. The formal artifact, rather than the language-model prose, is the machine-checked source for the certified statements.

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